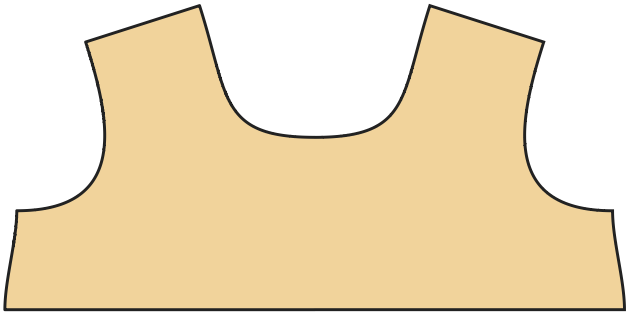
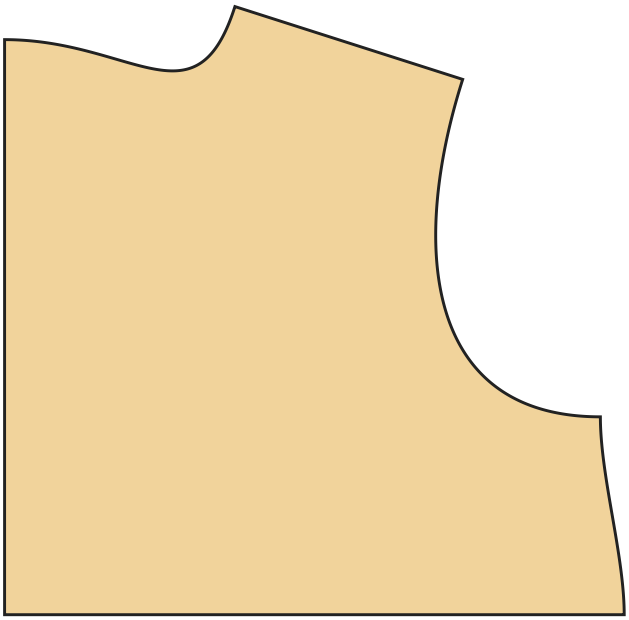
# DRESS WITH GATHERED SKIRT

## doll 45 cm (made to measure)

wide fit, Woven (0 % stretch)



Front view



Back view

| | |
|---|---|
| Size | **doll 45 cm (made to measure)** |
| Main fabric | **0.25 m at 150 cm width** |
| Notions | **Velcro** |
| Sewn with | **Sewing machine** |
| Seam allowance | **included: 0.6 cm · hem 0.6 cm, 1.0 cm** |
| Parts and steps | **6 parts · 24 steps** |
| Difficulty | **advanced · about 3.5 h** |

Read the steps once before you start. Every picture is drawn from your pattern pieces. Sew a test garment from cheap fabric first and adjust before you cut the good fabric.

# Parts



**A** 1×
Front · cut on fold



**B** 2×
Back



**C** 2×
Sleeve



**D** 1×
Neck binding (bias)



**E** 1×
Skirt front · cut on fold



**F** 2×
Skirt back

# Fabric and notions

**Main fabric**
0.25 m at 150 cm width

**Velcro**
The back opens completely: close it with narrow Velcro (1 cm) or 2–3 small snaps.

**Pins or clips**
Pin across the seam line.

**Polyester sewing thread**
In the colour of the fabric.

# Colours and lines

**Main fabric**
right side · wrong side

Sew now (machine)

Sew by hand

Fold / press line

Marking (chalk)

Pin

Notch: match them

Right/wrong side of the fabric = the side of the fabric. Left/right (as worn) = seen from the child wearing it: the left front leg is on the right when you look at the child.

# Machines and tools

### Sewing machine

straight stitch 2.5 mm; wide zigzag 4 × 1.5 mm over the edge; straight stitch 3 mm; 2 rows of the longest straight stitch, loose ends

### Iron

steam, cotton setting

### Pins or clips

### Fabric scissors

### Tape measure

### Chalk or a fabric pen

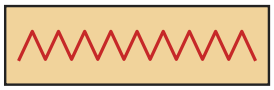
Steps are written for a sewing machine with zigzag. Have an overlocker or a coverstitch machine? Tell the generator which machines you own and the steps use them.

## Basics: the sewing machine



### Straight stitch

Seams in woven fabric, 2.5 mm; backstitch at the start and the end.



### Zigzag

Knit seams (narrow) and finishing the edges (wide).

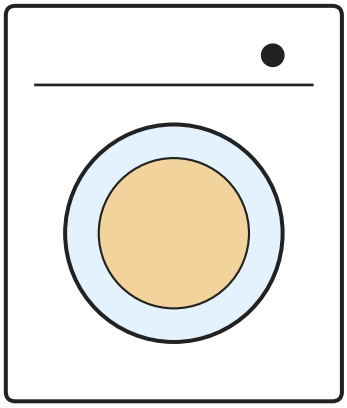


### Topstitching

Right side up, with the same distance to the edge all the way; use the edge of the presser foot as a guide.

PREPARE

## 1 Pre-wash the fabric



30–40°

1. Wash and dry it the way you will wash the garment later, so it shrinks now, not after sewing.

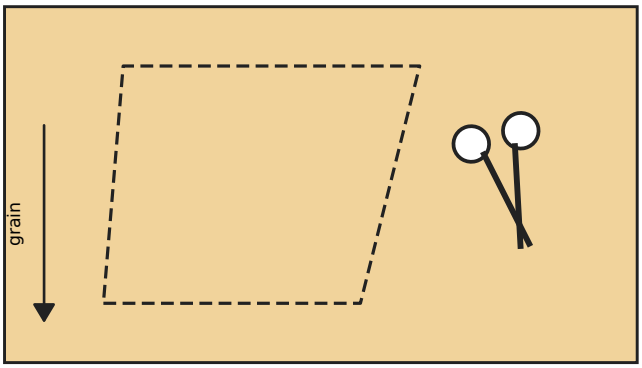
## 2 Cut out all pieces



Scissors

1. Follow the cutting layout
2. Grainline arrows parallel to the selvage.

> ✓ **Tip**
> Cut both layers at once with long strokes; keep the fabric flat on the table.

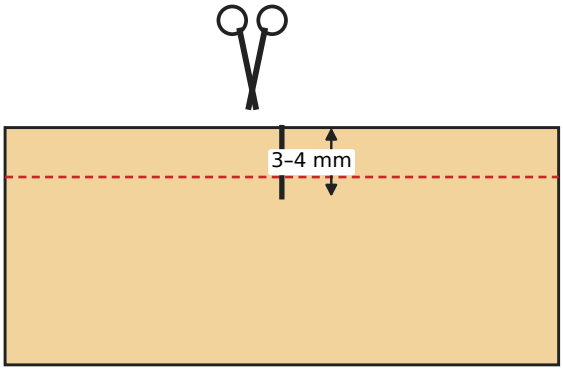


## 3 Snip the notches

Chalk

1. Cut 3–4 mm into the seam allowance at every notch.

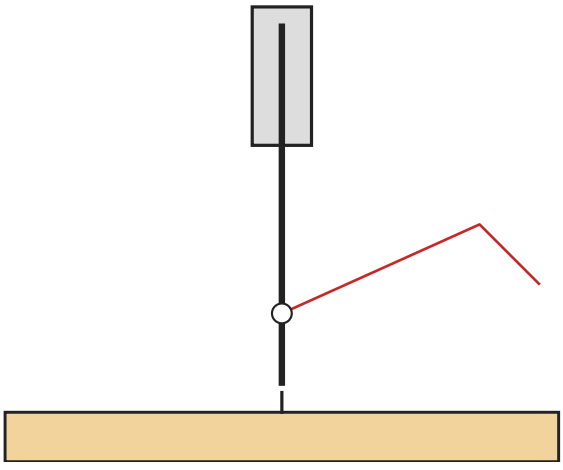
⚠ **Important**
Never deeper than half the seam allowance.




## 4 Set up the machine

1. Universal needle 80, polyester thread.
2. Sew a test seam on a scrap first.

✓ **Tip**
See the basics page at the front of the booklet.



SHOULDERS

## 5 Sew the shoulder seams

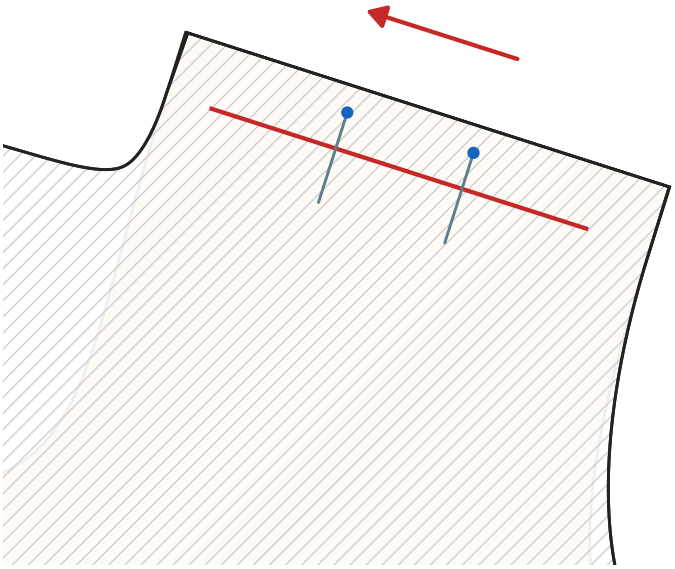
×2 both sides

Sewing machine

Pins

1. Front and back right sides together.

Sewing machine: straight stitch 2.5 mm



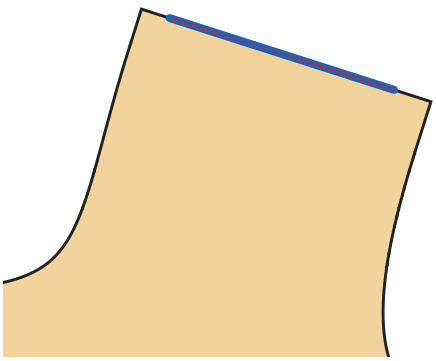
## 6 Finish the shoulder seams

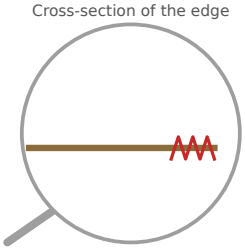
×2 both sides

Sewing machine

1. Both allowances together
2. Press them towards the back.

Sewing machine: wide zigzag 4 × 1.5 mm over the edge





BACK OPENING

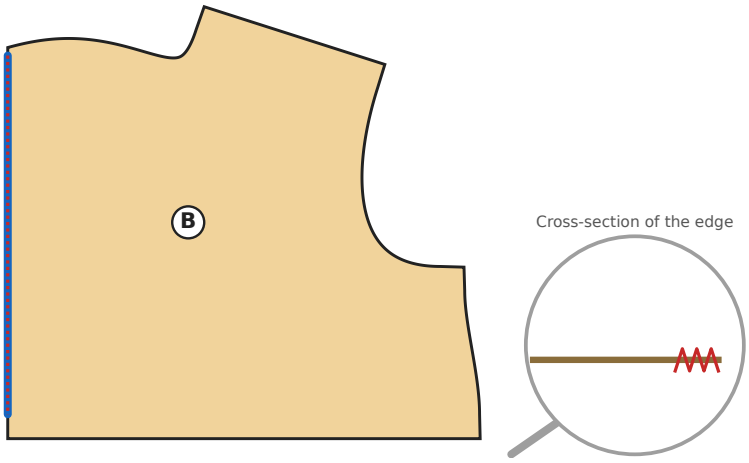
## 7 Finish the opening edges

×2
both sides

Sewing machine

1. Both centre back edges.

Sewing machine: wide zigzag 4 × 1.5 mm over the edge



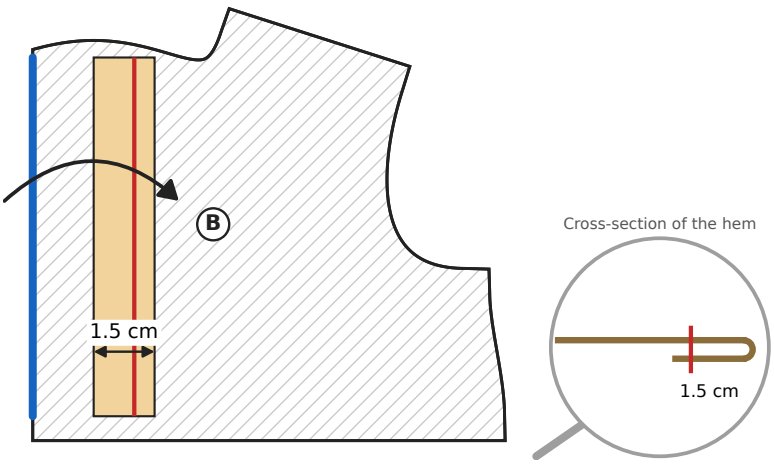

## 8 Turn in the opening edges

×2
both sides

Sewing machine

1. Press 1.5 cm to the inside and stitch it down.

Sewing machine: straight stitch 3 mm



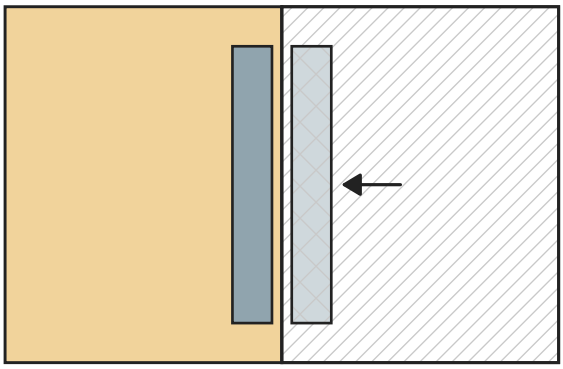
## 9 Sew on the Velcro

Sewing machine

1. The edges overlap at the centre back: soft side on the left back, hook side on the right back (or 2–3 snaps).

Sewing machine: straight stitch 3 mm

⚠ **Important**
The garment opens completely: easy dressing.



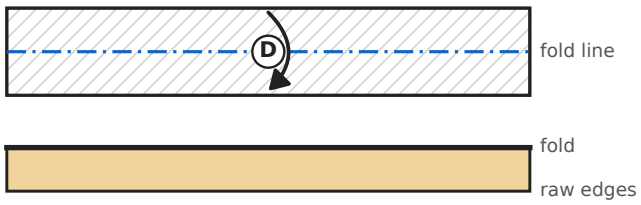
NECKLINE

## 10 Fold the binding in half

Iron

1. Lengthwise, wrong sides together
2. Press.

Iron: steam, cotton setting



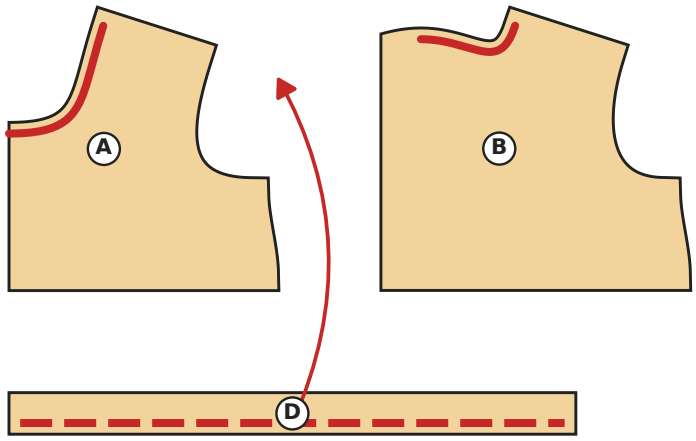
## 11 Sew the neck binding on

Sewing machine

Pins

1. Raw edges to the neckline, right sides together
2. Let each end stick out past the opening edge by the allowance.

Sewing machine: straight stitch 2.5 mm

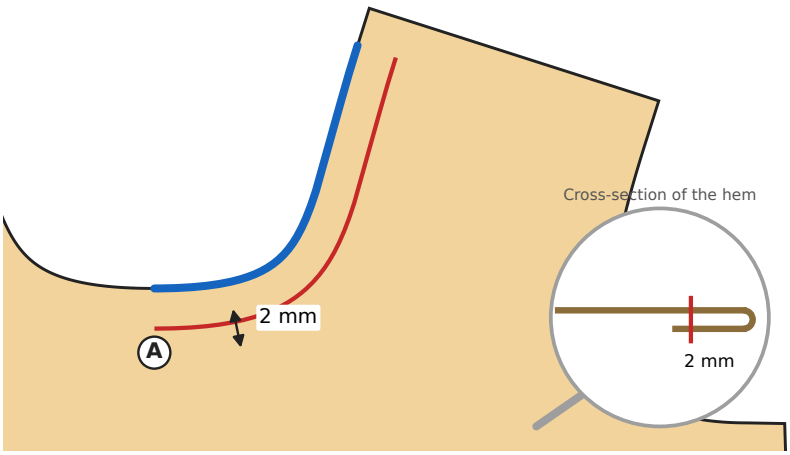



## 12 Stitch the binding down

Sewing machine

1. Fold the ends in, turn the binding to the inside over the allowance and stitch close to the fold.

Sewing machine: straight stitch 3 mm



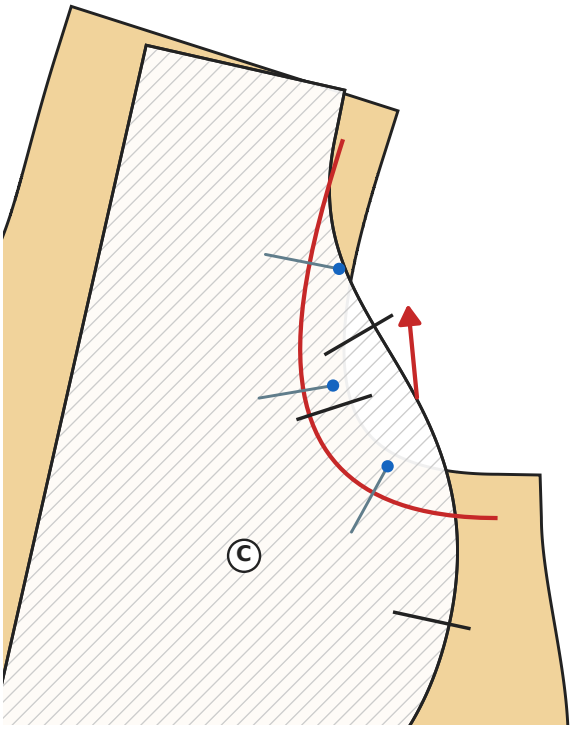
SLEEVES

## 13 Set in the sleeves

×2
both sleeves

Sewing machine

Pins




1. Top notch of the sleeve cap on the shoulder seam, front/back notches meet
2. Sew the sleeve in flat.

Sewing machine: straight stitch 2.5 mm

⚠ **Important**
Sleeve on the bottom: the machine eases the cap in.

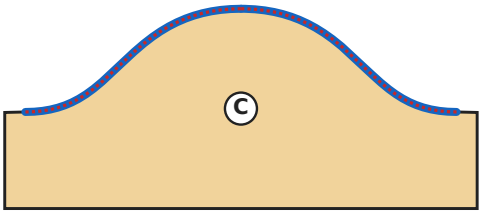
## 14 Finish the armhole seams

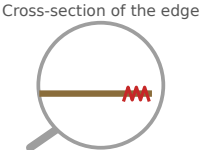
×2
both sleeves

Sewing machine

1. Both allowances together
2. Press them towards the back.

Sewing machine: wide zigzag 4 × 1.5 mm over the edge



SIDES

## 15 Sew side and sleeve seams

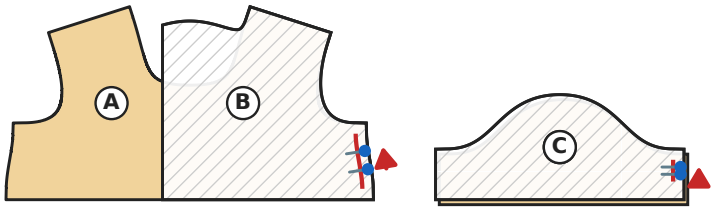
×2
both sleeves

Sewing machine

Pins

1. In one go from the hem over the underarm to the sleeve hem
2. The underarm seams meet exactly.

Sewing machine: straight stitch 2.5 mm



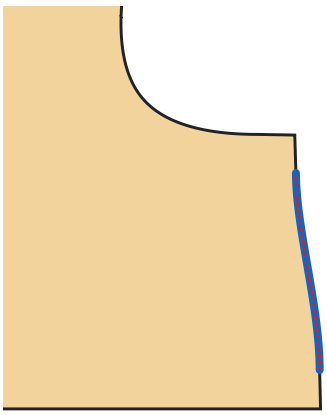

## 16 Finish the side seams

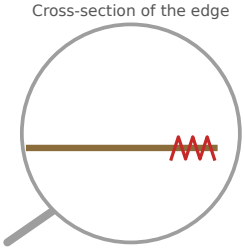
×2
both sides

Sewing machine

1. Both allowances together
2. Press them towards the back.

Sewing machine: wide zigzag 4 × 1.5 mm over the edge





SLEEVE HEMS

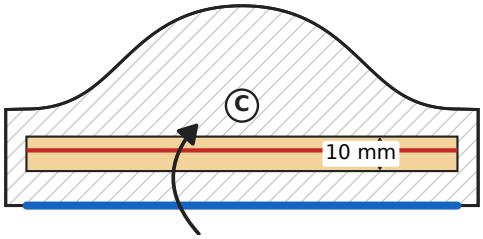
## 17 Hem the sleeves

×2
both sleeves

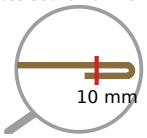
Sewing machine

1. Press 10 mm to the inside.

Sewing machine: straight stitch 3 mm




Cross-section of the hem




SKIRT

## 18 Sew the skirt centre back seam

Sewing machine

Pins

1. From the hem up to the notch
2. Above the notch it stays open (continues the back opening).

Sewing machine: straight stitch 2.5 mm



## 19 Sew the skirt side seams

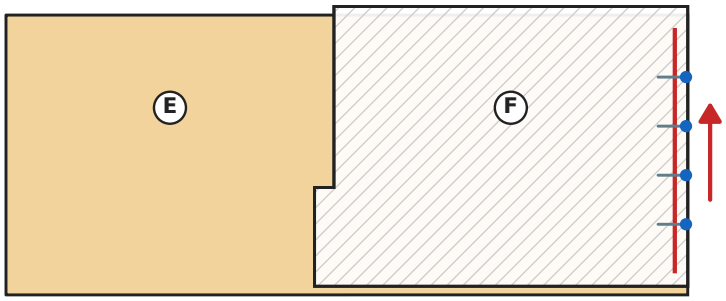
×2
both sides

Sewing machine

Pins

1. Skirt front and back right sides together.

Sewing machine: straight stitch 2.5 mm

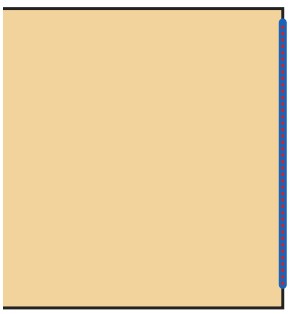


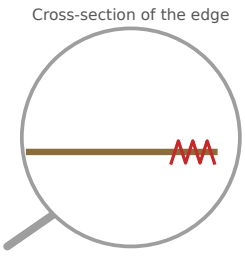

## 20 Finish the skirt side seams

×2
both sides

Sewing machine

Sewing machine: wide zigzag 4 × 1.5 mm over the edge



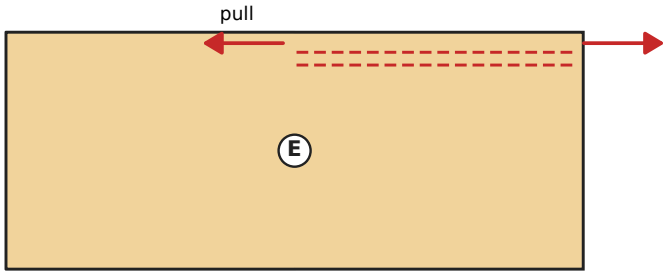


## 21 Gather the skirt top edge

Sewing machine

1. Two rows of long stitches
2. Pull them up to the bodice waist (gather ratio 1.6), spread the gathers evenly.

Sewing machine: 2 rows of the longest straight stitch, loose ends



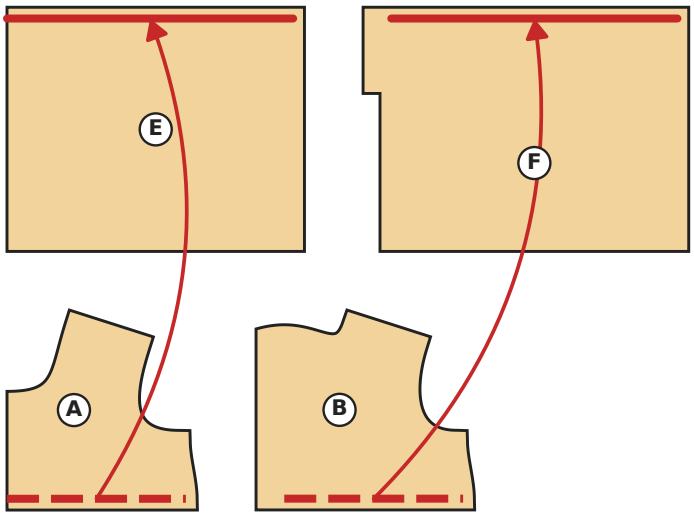

WAIST

## 22 Sew skirt and bodice together

Sewing machine Pins

1. Skirt right sides together to the bodice: side seams meet, centre front/back meet.

Sewing machine: straight stitch 2.5 mm



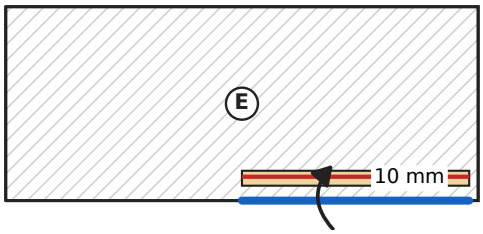
HEM

## 23 Sew the hem

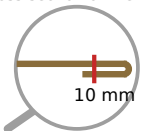
Sewing machine

1. Press the allowance under twice and stitch.

Sewing machine: straight stitch 3 mm




Cross-section of the hem



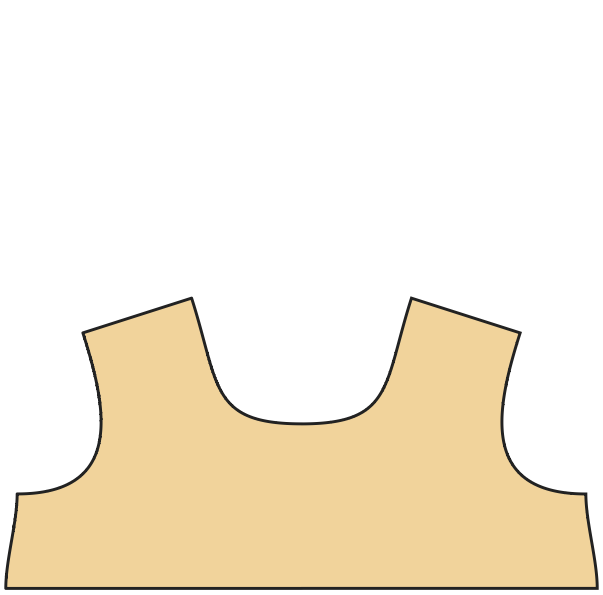

FINISH

## 24 Done!

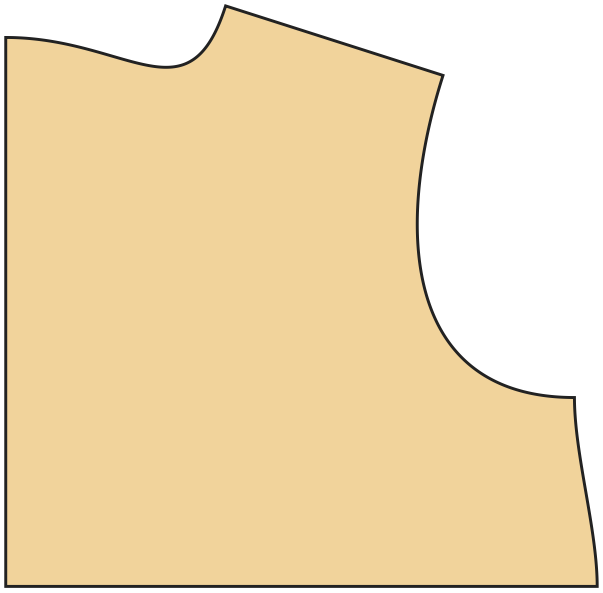
1. Press all seams.



# Done!



Front view



Back view

Press all seams.

✓ **After the test garment**
Try the test garment on the child: length, width at the hip and the waistband. Adjust the measurements or options and print the pattern again before you cut the good fabric.

# Dress with gathered skirt · doll 45 cm (made to measure)

wide fit, Woven (0 % stretch) · original parametric draft. The pattern is yours to use.

> **Important:** With a new pattern, first sew a **test garment** from inexpensive fabric with similar stretch and check the fit.

## Files

- PDF A4 tiles to glue (3 pages + overview): `doll_dress_45cm_a4_tiled.pdf`
- Preview A4 tiles: `doll_dress_45cm_a4_preview.png`
- SVG master 1:1: `doll_dress_45cm_master_1to1.svg`
- Cutting layout: `cutting_layout.png`
- Sewing instructions (PDF): `sewing_instructions.pdf`
- Sewing instructions (Markdown): `sewing_instructions.md`

## Printing

- Always print at **100 % / "Actual size"** (not "Fit to page").
- Measure the **5 × 5 cm** control square before you cut.
- Plotter (610 mm roll): the page is 600 mm wide in portrait. In the print dialog choose a custom paper size with exactly this width and length, and turn rotation/scaling off.

## Body measurements

| Body measurement | Value | Source |
|---|---|---|
| Body height | 45.0 cm | user |
| Chest | 30.0 cm | user |
| Waist | 28.0 cm | user |
| Hip | 31.0 cm | user |
| Head | 32.0 cm | user |
| Neck | 15.1 cm | derived |
| Inseam | 14.0 cm | user |
| Side length | 22.0 cm | user |
| Arm length | 12.0 cm | user |
| Back length | 9.0 cm | derived |
| Shoulder width | 3.2 cm | derived |
| Thigh | 18.0 cm | derived |
| Ankle | 11.4 cm | derived |

## Finished measurements (virtually sewn)

| Finished measurement | Value |
|---|---|
| Chest width | 36.9 cm |
| Hem width | 60.9 cm |
| Dress length from neck point | 20.7 cm |
| Sleeve length | 4.2 cm |
| Sleeve opening | 12.5 cm |
| Upper-arm width | 12.5 cm |
| Neckline circumference | 20.4 cm |
| Waist seam (bodice) | 38.1 cm |
| Skirt top edge (ungathered) | 60.9 cm |
| Skirt length | 11.4 cm |

## Pattern pieces

| Piece | Cutting | Fabric | Seam allowance |
|---|---|---|---|
| Front | cut 1 on fold | Main fabric | seam allowance 0.6 cm included; Neckline: 0.4 cm |
| Back | cut 2 (mirrored pair) | Main fabric | seam allowance 0.6 cm included; Back opening (turn under): 1.5 cm; Neckline: 0.4 cm |
| Sleeve | cut 2 (mirrored pair) | Main fabric | seam allowance 0.6 cm included; Sleeve hem: 1.0 cm |
| Neck binding (bias) | cut 1 | Main fabric | seam allowance 0.4 cm included |
| Skirt front | cut 1 on fold | Main fabric | seam allowance 0.6 cm included; Hem: 1.0 cm |
| Skirt back | cut 2 (mirrored pair) | Main fabric | seam allowance 0.6 cm included; Hem: 1.0 cm; Back opening (turn under): 1.5 cm |

## Fabric needed

- **Main fabric:** about 0.25 m at 150 cm width (single layer, incl. 10 % for shrinkage; folded would need 0.35 m)
- Doll clothes: narrow seam allowances (6 mm). Sew with a short stitch (1.5–2 mm) and finish woven edges with pinking shears, a narrow zigzag or a little fray stop.
- Thin, soft fabrics work best (cotton poplin, batiste, old T-shirts); thick fabric makes the tiny seams bulky.
- The back opens completely: close it with narrow Velcro (1 cm) or 2–3 small snaps.

Cutting layout: see `cutting_layout.png` (grainline arrow parallel to the selvage, "flipped" = place the piece face down, "×2" = cut through both layers of the folded fabric, which gives a mirrored pair, "(unfolded)" = cut-on-fold piece shown opened out).

**Cut on fold** (Front, Skirt front): the paper piece is only one half. Its fold edge (dash-dot line, bracket arrows "PLACE ON FOLD") is the centre of the piece: it gets no seam allowance and is never cut. Where the cutting layout shows a piece "(unfolded)" on a single layer, fold the fabric along the grain just wide enough for the piece and place the fold edge on that fold, or trace the paper piece, flip it over along the fold edge and trace the second half.

## Validation report

### Sewability check

| Area | Check | Target | Actual | Δ | Result | Note |
|---|---|---|---|---|---|---|
| Seam pairs | Side seam front ↔ back | 3.1 cm | 3.1 cm | +0.0 mm | □ OK | |
| Seam pairs | Armhole ↔ sleeve cap (ease) | ease 3–8 mm | 13.8 cm ↔ 14.3 cm | +5.4 mm | □ OK | Woven: 3–8 mm |
| Seam pairs | Shoulder seam front ↔ back | 3.7 cm | 3.7 cm | +0.0 mm | □ OK | |
| Seam pairs | Sleeve seam (front ↔ back) | 1.2 cm | 1.2 cm | +0.0 mm | □ OK | |
| Seam pairs | Neck binding ↔ neckline | 20.4 cm | 20.4 cm | +0.0 mm | □ OK | |
| Seam pairs | Waist seam bodice ↔ skirt (gathered) | 40–100 % of 60.9 cm | 38.1 cm = 63 % | -228.4 mm | □ OK | gather ratio 1.6 (allowed 1–2.5) |
| Seam pairs | Skirt side seam front ↔ back | 11.4 cm | 11.4 cm | +0.0 mm | □ OK | |
| Seam pairs | Skirt centre back seam (left ↔ right) | 7.4 cm | 7.4 cm | +0.0 mm | □ OK | open for the top 4.0 cm (notch) |
| Angles/transitions | Underarm: side seam ↔ armhole front | 90° | 91.9° | +1.9° | □ OK | |
| Angles/transitions | Underarm: side seam ↔ armhole back | 90° | 92.2° | +2.2° | □ OK | |

| Area | Check | Target | Actual | Δ | Result | Note |
|---|---|---|---|---|---|---|
| Angles/transitions | Shoulder ↔ armhole (front + back) | 180° | 180.6° | +0.6° | ✅ OK | |
| Angles/transitions | Sleeve cap top (smooth) | 180° | 178.2° | -1.8° | ✅ OK | |
| Angles/transitions | Sleeve underarm ↔ cap front | 90° | 90.9° | +0.9° | ✅ OK | |
| Angles/transitions | Shoulder ↔ neckline (front + back) | 180° | 180.9° | +0.9° | ✅ OK | |
| Angles/transitions | Neckline at centre front | 90° | 90.4° | +0.4° | ✅ OK | |
| Angles/transitions | Neckline at centre back | 90° | 89.0° | -1.0° | ✅ OK | |
| Angles/transitions | Hem at centre front | 90° | 90.0° | +0.0° | ✅ OK | |
| Angles/transitions | Hem ↔ side seam front | 90° | 88.8° | -1.2° | ✅ OK | |
| Angles/transitions | Skirt hem ↔ side seam | 90° | 90.0° | +0.0° | ✅ OK | |
| Angles/transitions | Skirt side seam ↔ waist | 90° | 90.0° | +0.0° | ✅ OK | |
| Angles/transitions | Skirt waist ↔ centre | 90° | 90.0° | +0.0° | ✅ OK | |
| Measurements | Chest circumference pattern = body + ease | 36.6 cm | 36.9 cm | +0.8 % | ✅ OK | body 30 cm + +22 % ease |
| Measurements | Head fits through the back opening | ≥ 33.6 cm | 38.1 cm | +45.0 mm | ✅ OK | neckline + both edges of the back opening ≥ head circumference + 5 % |
| Measurements | Dress goes on through the back opening | ≥ 32.6 cm | 46.0 cm | +134.9 mm | ✅ OK | waist seam + both edges of the skirt opening ≥ hip + 5 % |
| Measurements | Dress length plausible | 14.9 cm – 33.5 cm | 20.7 cm | +0.0 mm | ✅ OK | from the neck point |
| Geometry | Front: outline & seam allowance | closed, SA present | ok | | ✅ OK | seam allowance 0.6 cm included; Neckline: 0.4 cm |
| Labels | Front: grainline & cutting | present | cut 1 on fold | | ✅ OK | Main fabric |
| Geometry | Back: outline & seam allowance | closed, SA present | ok | | ✅ OK | seam allowance 0.6 cm included; Back opening (turn under): 1.5 cm; Neckline: 0.4 cm |
| Labels | Back: grainline & cutting | present | cut 2 (mirrored pair) | | ✅ OK | Main fabric |
| Geometry | Sleeve: outline & seam allowance | closed, SA present | ok | | ✅ OK | seam allowance 0.6 cm included; Sleeve hem: 1.0 cm |
| Labels | Sleeve: grainline & cutting | present | cut 2 (mirrored pair) | | ✅ OK | Main fabric |
| Geometry | Neck binding (bias): outline & seam allowance | closed, SA present | ok | | ✅ OK | seam allowance 0.4 cm included |
| Labels | Neck binding (bias): grainline & cutting | present | cut 1 | | ✅ OK | Main fabric |
| Geometry | Skirt front: outline & seam allowance | closed, SA present | ok | | ✅ OK | seam allowance 0.6 cm included; Hem: 1.0 cm |

| Area | Check | Target | Actual | Δ | Result | Note |
|---|---|---|---|---|---|---|
| Labels | Skirt front: grainline & cutting | present | cut 1 on fold | | □ OK | Main fabric |
| Geometry | Skirt back: outline & seam allowance | closed, SA present | ok | | □ OK | seam allowance 0.6 cm included; Hem: 1.0 cm; Back opening (turn under): 1.5 cm |
| Labels | Skirt back: grainline & cutting | present | cut 2 (mirrored pair) | | □ OK | Main fabric |
| Geometry | Back: turned allowance lies flat | ≤ 0.5 mm | 0.0 mm | | □ OK | Back opening (turn under): corners mirrored at the fold line |
| Geometry | Sleeve: turned allowance lies flat | ≤ 0.5 mm | 0.0 mm | | □ OK | Sleeve hem: corners mirrored at the fold line |
| Geometry | Front: cut-on-fold edge | straight, no SA, 90° corners, on grain | ok, corners ≤ 0.4° off 90° | | □ OK | Centre (fold) |
| Geometry | Skirt front: cut-on-fold edge | straight, no SA, 90° corners, on grain | ok, corners ≤ 0.0° off 90° | | □ OK | Centre (fold) |
| Labels | Front: notches | on the cutting line | 1 notch mark(s) | | □ OK | |
| Labels | Back: notches | on the cutting line | 2 notch mark(s) | | □ OK | |
| Labels | Sleeve: notches | on the cutting line | 4 notch mark(s) | | □ OK | 1 at corners |
| Labels | Skirt back: notches | on the cutting line | 2 notch mark(s) | | □ OK | |

**Result:** 45 OK, 0 warning(s), 0 error(s): the pattern is sewable.

## Construction values

Override any of these with the option `construction`, values in mm (e.g. `{"construction": {"chest_width": 366}}`).

| Name | Value | Derived from | Note |
|---|---|---|---|
| `chest_width` | 36.6 cm | chest × (1 + 22 % ease) | |
| `neck_width_half` | 3.5 cm | neck / 5 + 1.5 cm | |
| `front_neck_depth` | 4.0 cm | neck_width_half + 1.5 cm | |
| `back_neck_depth` | 0.5 cm | fixed | |
| `shoulder_length` | 3.7 cm | shoulder + allowance for fit 'wide' | |
| `shoulder_slope` | 17 | fixed | |
| `armhole_depth` | 6.3 cm | 18 % of chest_width + 2.5 cm + 3 %, limited to option armhole_depth_max | |
| `body_length` | 9.3 cm | option body_length | |

ting layout · Dress with gathered skirt doll 45 cm (made to measu

Main fabric: 150 cm wide, single layer
needed ≈ 0.25 m

